Class Task

Doubly Linked List

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Question

Design a double LinkedList having six nodes, print the first and the last node.

Solution

A doubly linked list is implemented with a Node class (data, prev, next) and a DoublyLinkedList class. Six nodes are appended, then the first and last node values are printed.

```
class Node:
    def __init__(self, data):
        self.data = data
        self.prev = None
        self.next = None


class DoublyLinkedList:
    def __init__(self):
        self.head = None
        self.tail = None

    def append(self, data):
        node = Node(data)
        if self.head is None:
            self.head = node
            self.tail = node
            return
        node.prev = self.tail
        self.tail.next = node
        self.tail = node

    def get_first(self):
        return self.head.data if self.head else None

    def get_last(self):
        return self.tail.data if self.tail else None

    def display(self):
        current = self.head
        values = []
        while current:
            values.append(str(current.data))
            current = current.next
        print(" <-> ".join(values) + " -> None")


def main():
```

```
    dll = DoublyLinkedList()

    # Create a doubly linked list having six nodes
    for value in [10, 20, 30, 40, 50, 60]:
        dll.append(value)

    print("Doubly Linked List:")
    dll.display()
    print()
    print("First node:", dll.get_first())
    print("Last node:", dll.get_last())


if __name__ == "__main__":
    main()
```

Program Output

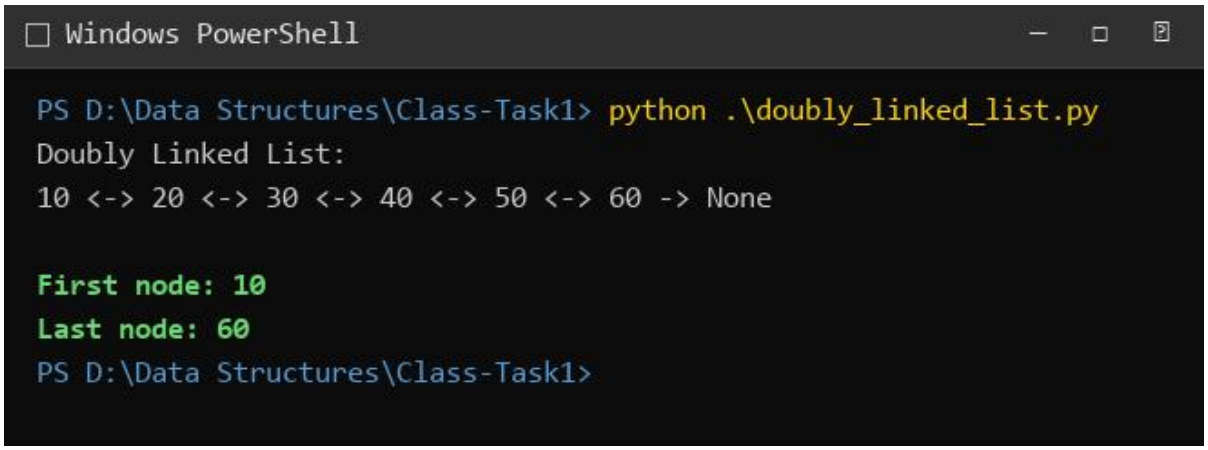


Figure 1: Running the program in Windows PowerShell